

Assignment 1: Polynomial Regression

Problems: var1 and var2

1. Objective

The aim of this assignment was to build polynomial regression models for the two given problems and use them to predict the target value y for the corresponding test datasets. The models were built using the feature and polynomial-degree settings specified in the assignment.

The final models were trained on the complete training datasets and then used to generate predictions for the test datasets. The assignment evaluates the predictions using Mean Squared Error (MSE) and R^2 against the hidden test targets.

2. Model Selection

var1: The problem is based on steam-turbine optimisation. The assignment specifies an optimal polynomial of degree 3 using the first three features. Therefore, **x1, x2 and x3** were selected, and polynomial terms up to total degree 3 were generated.

var2: The problem is based on subterranean thermal-reservoir mapping. The assignment specifies an optimal polynomial of degree 4 using only the first feature. Therefore, only **x1** was selected, with polynomial terms up to degree 4.

Problem	Features	Degree	Polynomial terms
var1	x1, x2, x3	3	19
var2	x1	4	4

3. Data and Preprocessing

Each training dataset contains 1,000 observations and each test dataset contains 1,000 observations. The training files contain the target column y , while the test files contain only the input features. The test target values were not used during model training.

For each problem, only the features required by the assignment were selected. The selected numerical features were passed directly to the polynomial transformation.

4. Methodology

The models were implemented using scikit-learn. First, PolynomialFeatures was used to expand the selected features to the required degree. LinearRegression was then fitted to the expanded training data. The same polynomial transformation was applied to the corresponding test data before prediction.

Pipeline: selected features → polynomial feature expansion → linear regression → predicted y

5. Final Training and Predictions

Each model was trained using all 1,000 available training observations. The trained models were then used to predict the 1,000 rows in their corresponding test datasets.

The generated prediction files contain a single column named **y** and do not contain an additional index column. This follows the format required by the sample submission.

Generated prediction files

- BT2024155_pred_var1.csv — 1,000 predictions
- BT2024155_pred_var2.csv — 1,000 predictions

6. Reproducibility

The GitHub repository contains the training and inference code, input datasets, prediction files, requirements file and this report. The main implementation is in **src/train_and_predict.py**.

The project can be reproduced by installing the packages listed in requirements.txt and running the training and inference script from the project root.

GitHub Repository: <https://github.com/DhanushDev311/BT2024155-Polynomial-Regression>

7. Conclusion

Both polynomial regression tasks were completed using the configurations specified in the assignment. For var1, the model uses the first three features with degree 3. For var2, the model uses only the first feature with degree 4. The final models were trained on the complete training datasets and used to produce the required test predictions.

The source code and generated prediction files are included in the GitHub repository.